

How the Proof Works

For the global three-dimensional Navier–Stokes regularity problem, the fluid obeys

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0.$$

For a smooth solution that decays at infinity, taking the L^2 inner product with u gives

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2.$$

This bounds the kinetic energy at each time and the accumulated gradient over the interval. It does not bound $\|\nabla u(t)\|_2$ at each individual time.

As a vortex tube stretches along its axis, the parcels of rotating fluid inside it lengthen in that direction. Incompressibility preserves each parcel's volume, so the axial extension is balanced by contraction across the tube. The rotation is now carried by a thinner core. With $\omega = \nabla \times u$, taking the curl of the equation gives

$$D_t \omega := \partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The term $(\omega \cdot \nabla)u$ is vortex stretching. When the surrounding strain pulls in the direction of ω , this term increases $|\omega|$: the vorticity becomes stronger as the tube grows longer and thinner. Since ω is built from first spatial derivatives of u , the velocity changes more sharply across the core. Viscosity acts against that sharpening through $\nu \Delta \omega$.

To follow this balance into a smaller core, we need a precise way to shrink the solution without changing the equation it solves. Navier–Stokes scaling builds a narrower, faster field from (u, p) :

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t), \quad \lambda > 1.$$

If the original structure has width r , velocity scale U , and duration τ , its rescaled copy has width r/λ , velocity scale λU , and duration τ/λ^2 . Substitution gives

$$\begin{aligned} & \partial_t u_\lambda + (u_\lambda \cdot \nabla) u_\lambda + \nabla p_\lambda - \nu \Delta u_\lambda \\ &= \lambda^3 [\partial_t u + (u \cdot \nabla) u + \nabla p - \nu \Delta u](\lambda x, \lambda^2 t) = 0. \end{aligned}$$

The common factor λ^3 cancels, so the rescaled fields obey the original equation. Narrowing the core strengthens nonlinear sharpening and viscous diffusion by an identical factor; scale alone gives viscosity no new advantage.

The narrower field has velocity scale λU , but it occupies λ^{-3} times the volume. Its kinetic energy scales as

$$\|u_\lambda(\cdot, t)\|_2^2 = \lambda^{-1} \|u(\cdot, \lambda^2 t)\|_2^2.$$

The energy norm becomes smaller as the core becomes thinner, even while the gradients become larger.

The proof needs a norm that does not shrink as the core narrows. For the homogeneous Sobolev norms,

$$\widehat{u_\lambda}(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t),$$

and a change of variables gives

$$\begin{aligned} \|u_\lambda(\cdot, t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} |\widehat{u}(\xi/\lambda, \lambda^2 t)|^2 d\xi \\ &= \lambda^{2s-1} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}^2. \end{aligned}$$

At $s = \frac{1}{2}$, the factor is $\lambda^0 = 1$. Contracting the field changes its spatial scale without changing its $\dot{H}^{1/2}$ size. Below that height, concentration makes the norm smaller; above it, concentration makes the norm larger. The energy norm becomes smaller under concentration; the critical norm does not. That is why the proof begins at $s = \frac{1}{2}$.

The factor λ^2 in time tells us how long each smaller turn would take. Suppose one evolving vortex repeats the full Navier–Stokes rescaling from turn to turn, with its width contracted by a fixed factor $0 < \rho < 1$. Then

$$r_j = \rho^j r_0, \quad U_j = \rho^{-j} U_0, \quad \tau_j := \frac{r_j}{U_j} = \rho^{2j} \tau_0.$$

The duration of each turn is ρ^2 times the preceding turn, and

$$\sum_{j=0}^{\infty} \tau_j = \tau_0 \sum_{j=0}^{\infty} \rho^{2j} = \frac{\tau_0}{1 - \rho^2} < \infty.$$

Under this repeated-contraction hypothesis, infinitely many turns fit before a finite time while U_j diverges. This is the Zeno cascade: a thinner and faster vortex at every turn, with nonlinear sharpening and viscous diffusion still scaled together. Finite energy and finite duration do not, by themselves, stop the sequence.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

What you'd expect to see in this pattern, if a singularity were to form, as you go up the time derivatives, you'd expect to see less spatial regularity, not more.

The equation moves in the other direction. Its first three time derivatives are

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The pattern is

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

Set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p,$$

Taking the divergence and using incompressibility gives the pressure at that order:

$$-\Delta Q_n = \sum_{a=0}^n \binom{n}{a} \partial_i \partial_j ((V_a)_i (V_{n-a})_j).$$

The line to keep in view is $V_{n+1} = \nu \Delta V_n - \dots$. Moving one step up in time brings in two spatial

derivatives of the response below it. Transport and pressure stay attached, but viscosity supplies the link from temporal order to spatial Sobolev order.

At the critical height, set

$$\Lambda = (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

If $\mathcal{P}_s$ denotes the complete transport–pressure contribution at height s , the energy identity is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

Starting from $H = E_{1/2}$ and repeating that identity gives

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The first term has reached spatial height $n + \frac{1}{2}$; the sum carries every transport–pressure term generated on the way. This is the opposite of the blow-up picture. Rising through that temporal sequence was supposed to reveal less spatial regularity. The equation exposes higher Sobolev height instead. As you do that, you get higher order Sobolev embeddings:

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3), \quad \bigcap_{s < \infty} H^s(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Seeing a higher energy in the identity is not enough. It has to remain finite all the way to T_* . Fix smooth divergence-free finite-energy data u_0 , viscosity $\nu > 0$, and its maximal pressure-complete solution (u, p) on $[0, T_*)$. Assume $T_* < \infty$. For finite N and M , collect the adjacent rows in

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^N \left(\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right).$$

Adding the corresponding Q_n and ∇Q_n coordinates gives the finite pressure-complete rectangle $\mathcal{R}_{N,M}[u_0; T]$. The manuscript’s whole-terminal theorem bounds each adjacent pair on the full interval:

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_{n+1} \in L^2(0, T_*; H^{M-1}), \quad 0 \leq n \leq N.$$

Restriction to $(0, T)$ can only decrease those norms, and there are only finitely many rows, so

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

The bound may depend on N and M . No uniform constant over all orders is needed. At $n = 0$, for every finite m , the adjacent pair gives

$$u \in L^2(0, T_*; H^{m+1}), \quad \partial_t u \in L^2(0, T_*; H^{m-1}),$$

The Lions–Magenes trace theorem applied to $H^{m+1} \hookrightarrow H^m \hookrightarrow H^{m-1}$, gives a unique trace at the endpoint:

$$u \in C([0, T_*]; H^m) \quad \text{for every finite } m.$$

Increasing N or M does not create a new solution. Every rectangle is a finite part of the maximal history from u_0 , so overlapping rectangles agree on every shared coordinate. Their projective limit is

$$\mathcal{I}[u_0] := \varprojlim_{N, M < \infty} \mathcal{R}_{N,M}[u_0; T_*], \quad u \in \bigcap_{r, m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

The intersection is not one bound over infinitely many orders. Each coordinate was bounded at a finite (N, M) ; compatibility says that all of those finite bounds belong to one velocity–pressure history.

The zeroth temporal row, $\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0]$, now has one terminal value at every finite spatial height:

$$u_* := u(T_*) \in \bigcap_{m < \infty} H_\sigma^m(\mathbb{R}^3) = H_\sigma^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The recurrence and pressure equation recover every finite coordinate of the terminal jet from u_* . The uniform H^3 row gives a local existence time $\delta_0 > 0$ that works for u_* and for every sufficiently late slice $u(t_0)$. Choose $t_0 < T_*$ with $T_* - t_0 < \delta_0/2$, and restart from $u(t_0)$. Uniqueness makes this restart identical to the original solution on $[t_0, T_*)$, so it reaches T_* with value u_* . To the right of T_* , uniqueness identifies it with the solution restarted from u_* . The two restarts join into one smooth pressure-complete solution beyond T_* , contradicting maximality. Thus $T_* = \infty$.

Once $V_r \in C_t H_x^m$ is available for every finite r and m , any finite downstream estimate comes from choosing

$$m > k + \frac{3}{2} - \frac{3}{p}, \quad 2 \leq p \leq \infty,$$

which gives, on every finite time interval,

$$V_r \in C_t H_x^m \hookrightarrow L_t^q W_x^{k,p}, \quad 1 \leq q \leq \infty.$$

$$u_0 \longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty.$$